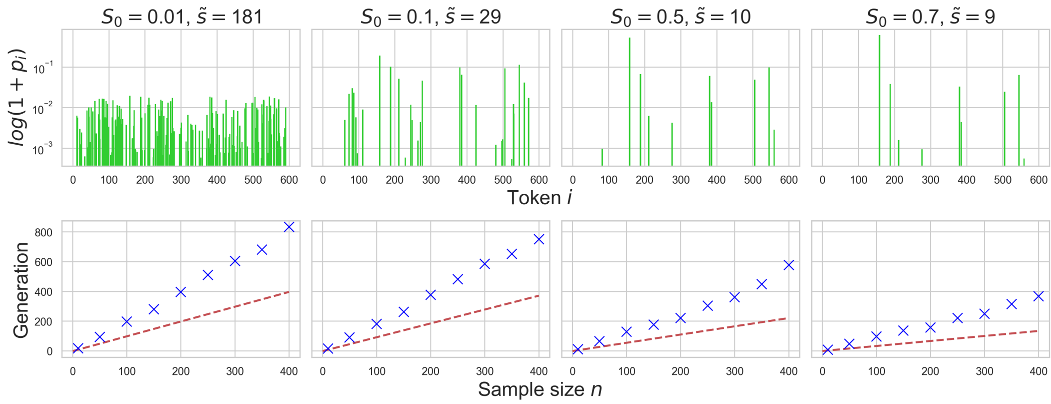

Figure 2: **Fully Synthetic case for different initial distribution $\boldsymbol{p}^{(0)}$.** Total collapse time is plotted as a function of the initial distribution $\boldsymbol{p}$ and the sample size n. **(Top)** The initial distribution $\boldsymbol{p}$ with different values of S_0 and support size $\tilde{s}$. The x-axis represents tokens $i \in \{1, 2, \ldots, 600\}$ while the y-axis represents the probabilities in log scale. **(Bottom)** Each cross represents the average total collapse time over 100 runs for a particular sample sizes $n \in \{10, 50, 100, 150, \ldots, 400\}$. The red dashed line depicts the lower bound on $\mathbb{E}T$ given by (8).

$p_j^{(m)}$ of outputting unlikely tokens j (those j's for which p_j is small) will decrease as $\boldsymbol{p}^{(m)}$ converges to some Dirac mass δ_i. After many generations, all but one $p_j^{(m)}$ will go to 0, exhibiting late model collapse where all the randomness of the original distribution is lost.

3.2 *Partially Synthetic:* Handling model collapse with real data

As described in the previous section, total collapse is unavoidable when training solely on synthetic data. In this section, we consider the case in which real data is incorporated at each generation. In this case, $\boldsymbol{p}^{(m)}$ can be simply seen as a weighted average of $\boldsymbol{p}^{(1)}$ and an estimate of $\boldsymbol{p}^{(m-1)}$ with n samples. We also assume that $\boldsymbol{p}^{(1)}$ is nontrivial, thereby implying that, on average, $\boldsymbol{p}^{(m)}$ will not be a Dirac mass. In what follows, we quantify the variability in the data distribution across different generations, as well as the distribution drift $\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ from the first-generation model. We particularly show that collapse can be avoided if enough real data is injected into the recursive training process.

Theorem 2 (Model Variation)**.** *Consider the Partially Synthetic setting. For $m \geq 1$ we have*

$$S_{m+1} = \frac{\frac{1}{N}\left[1 + 2\alpha - \left(1 - \frac{1}{N}\right)\alpha\beta^m\right]}{1 + (1 + 1/N)\alpha} + \frac{(1 - \frac{1}{N})S_0}{1 + (1 + 1/N)\alpha}\left[1 + \alpha - \frac{\alpha\beta^m}{N}\right], \tag{9}$$

where $\alpha := \frac{n}{N+n}$ and $\beta := \alpha\left[(1 + \frac{1}{N})\alpha - \frac{1}{N}\right]$.

Theorem 2 provides a control of the variance S_m in the *Partially Synthetic* setting. Essentially, when $n \ll N$, we have $S_m \approx \frac{1}{N} + (1 - \frac{1}{N})S_0 \approx S_0$, which is not surprising since the training data for each generation model are dominated by real data in this case. However, even when the number of synthetic data is much larger than the original dataset (i.e. $n \gg N$), we have $\alpha \approx 1 \approx \beta$ and hence $S_{m+1} \approx 1/N + (2 + 1/N)^{-1}(1 - 1/N)(2 - 1/N)S_0$, which approaches S_0 for large N.

To further refine our analysis, we present a result that directly controls the deviation $\mathbb{E}\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ between the conditional distributions from first and m-th generations. This allows us to have a quantitative control over the distribution shift. When n is sufficiently small, we have a sharper control over this deviation by exploiting the concentration results

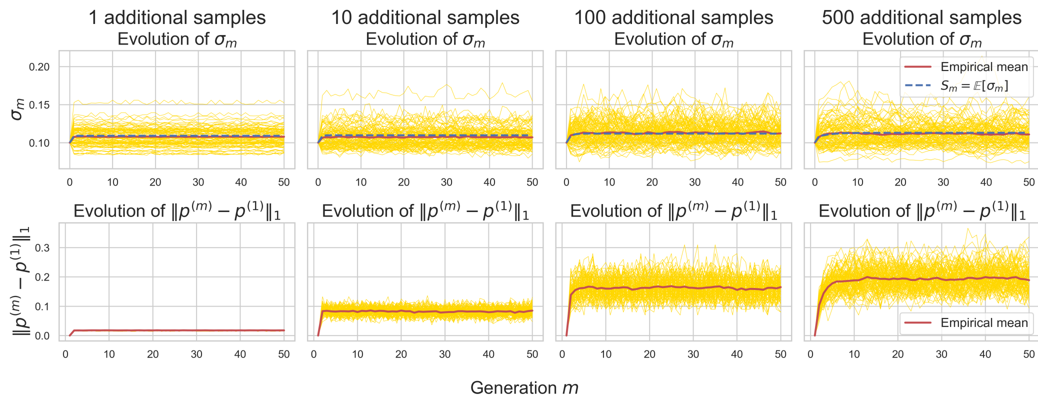

Figure 3: **Partially Synthetic case with different sample sizes n.** A hundred experiments were run for 50 generations for $N = 100$ and different values of n. Each yellow line represents the evolution of σ_m (top row) or $\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ (bottom row) in one experiment, with the red line being the empirical mean across 100 runs. The blue dashed lines plot the formula for S_m given by (9). The initial distribution $\boldsymbol{p}$ satisfies $s = 600$, $\tilde{s} = 52$, and $S_0 = 0.1$.

from (Mardia et al., 2020). Essentially, this allows us to estimate the maximum number of synthetic samples n to ensure that the distribution $\boldsymbol{p}^{(m)}$ stays close to $\boldsymbol{p}^{(1)}$.

Theorem 3 (Model Deviation). *Consider the Partially Synthetic setting and define*

$$G_n(s) := \begin{cases} C_1 s e^{\frac{C_0 n}{2e}} & \text{if } \frac{C_0}{e}n + 2 \le s; \\ C_1 s \left(\frac{C_0 n}{s}\right)^{s/2} & \text{if } \frac{C_0}{4}n + 2 \le s < \frac{C_0}{e}n + 2; \\ (2^s - 2) & \text{if } s < \frac{C_0}{4}n + 2, \end{cases} \tag{10}$$

where $C_0 = \frac{e^3}{2\pi} \approx 3.19$ and $C_1 = \frac{6e}{\pi^{3/2}} \approx 2.93$, and s is the vocabulary size. Then, for $m \ge 2$,

$$\mathbb{E}\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1 < \frac{1}{N}\sqrt{\frac{\pi n}{2}} G_n(s).$$

We point out that the upper bound on the deviation $\mathbb{E}\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ is independent of the generation m. Since the deviation from $\boldsymbol{p}^{(0)}$ to $\boldsymbol{p}^{(1)}$ is inevitable and independent of n, we give the result in $\mathbb{E}\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$, from which $\mathbb{E}\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(0)}\|_1$ can be estimated by the triangle inequality. Theorem 3 allows us to estimate the maximum number of synthetic samples n that can be used if we want $\boldsymbol{p}^{(m)}$ to stay ϵ-close to $\boldsymbol{p}^{(1)}$ in L^1 norm. For example, when $C_0 n/e + 2 \le s$, for any $\epsilon > 0$ we can take

$$n \le 2\pi e^{-2} \min\left[s - 2, \log\left(\frac{\sqrt{2}\pi N\epsilon}{6es}\right)\right] \tag{11}$$

in order for $\mathbb{E}\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ to be less than ϵ. In other words, to ensure small L^1 deviation n should be taken to be logarithmic in the ratio $N\epsilon/s$, which highlights that the amount of synthetic data should be exponentially smaller compared to real data in order to ensure that $\boldsymbol{p}^{(m)}$ remains close to $\boldsymbol{p}^{(1)}$. We show through simulations the effect of the sample size n and the initial distribution $\boldsymbol{p}$. In Figure 3, we show that if we fix the initial distribution and increase the amount of synthetic data n, the dispersion σ_m stays relatively constant while the distribution drift $\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ increases. In contrast, when we fix n and increase the values of S_0, σ_m increases but $\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ actually decreases as depicted in Figure 4. This behavior can be explained by Theorem 4 in Appendix B, which is more general than Theorem 3 in the sense that it captures the dependence of $\mathbb{E}\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ on the randomness of $\boldsymbol{p}$ and hence provides a sharper bound on the expected deviation.

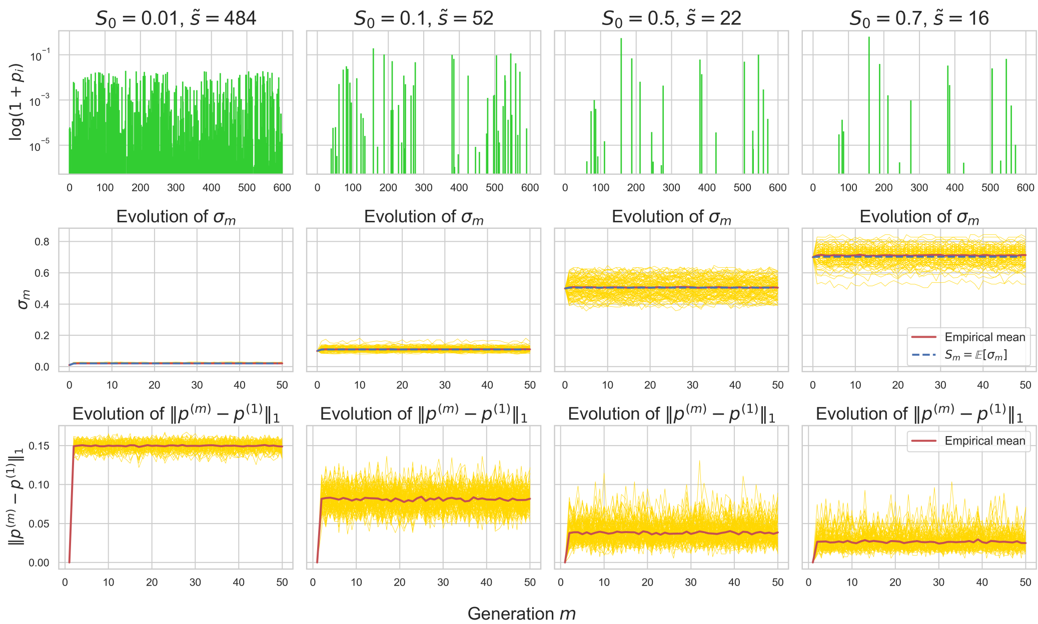

Figure 4: **Partially Synthetic case with different initial distributions $\boldsymbol{p}$.** A hundred experiments were run for 50 generations for $n = 10$, $N = 100$ and different initial distributions $\boldsymbol{p}$ shown in the top row. Notice that as S_0 increases, the deviation $\|p^{(m)} - p^{(1)}\|_1$ decreases, as suggest by the inequality (19).

4 Experiments

4.1 Transformer-based Models

To support our findings in a realistic setting, we conduct experiments with a decoder-only generative model for text generation. For this model, we consider the aforementioned *Fully Synthetic* and *Partially Synthetic* settings using the model parameters in Appendix C. We consider a simple character-level tokenizer using the tiny Shakespeare dataset[3] yielding a vocabulary of size $s = 65$. We first train a model for 2000 iterations on this dataset and consider the trained model as the ground-truth distribution $\boldsymbol{p}^{(0)}$. The next-generation models are trained recursively using the exact same architecture and training parameters. This setting allows us to reduce the effect of functional approximation error thereby focusing only on the effect of statistical approximation error.

The results of these experiments are summarized in Figure 5. The top left plot therein depicts the deviation $\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ which is averaged over 300 random contexts generated by the ground-truth generative model $\boldsymbol{p}^{(0)}$. From this plot, we clearly see that $\|\boldsymbol{p}^{(m)} - \boldsymbol{p}^{(1)}\|_1$ diverges over generations in the *Fully Synthetic* setting, while mixing with real data ensures stability over generations.

The effect of synthetic data is further noticed in the validation loss of the next-generation models where we see an overfitting effect in the *Fully Synthetic* setting. We point out that this effect might also be associated with the functional approximation error and was rigorously studied by Dohmatob et al. (2024a) in the case of linear regression, where the authors have shown that synthetic data affect the usual scaling laws. We believe that similar conclusions can be obtained with our statistical model by incorporating the functional approximation error, this can be achieved for instance by supposing that the context embeddings $\boldsymbol{e}_i$'s are

[3] `https://raw.githubusercontent.com/karpathy/char-rnn/master/data/tinyshakespeare/input.txt`